lab1.2.go

```go
package main

import (
    "bufio"
    "fmt"
    "os"
    "strconv"
    "strings"
)

const (
    minVal = -1000000
    maxVal = 1000000
)

var reader = bufio.NewReader(os.Stdin)

func readLine(prompt string) string {
    fmt.Print(prompt)
    text, _ := reader.ReadString('\n')
    return strings.TrimSpace(text)
}

func readInt(prompt string) int {
    for {
        n, err := strconv.Atoi(readLine(prompt))
        if err != nil {
            fmt.Println("Invalid input: enter a whole number.")
        } else if n < minVal || n > maxVal {
            fmt.Printf("Out of range: enter a value between %d and %d.\n", minVal, max
        } else {
            return n
        }
    }
}

func readFloat(prompt string) float64 {
    for {
        f, err := strconv.ParseFloat(readLine(prompt), 64)
        if err != nil {
            fmt.Println("Invalid input: enter a number.")
        } else if f < minVal || f > maxVal {
            fmt.Printf("Out of range: enter a value between %d and %d.\n", minVal, max
        } else {
            return f
        }
    }
}

func main() {
    for {
```

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```
52          fmt.Println("\n--- Integer calculator ---")
53          a := readInt("Enter first integer: ")
54          b := readInt("Enter second integer: ")
55          fmt.Printf("%d + %d = %d\n", a, b, a+b)
56          fmt.Printf("%d - %d = %d\n", a, b, a-b)
57          fmt.Printf("%d * %d = %d\n", a, b, a*b)
58
59          fmt.Println("\n--- Floating-point calculator ---")
60          x := readFloat("Enter first number: ")
61          y := readFloat("Enter second number: ")
62          fmt.Printf("%.2f + %.2f = %.2f\n", x, y, x+y)
63          fmt.Printf("%.2f - %.2f = %.2f\n", x, y, x-y)
64          fmt.Printf("%.2f * %.2f = %.2f\n", x, y, x*y)
65
66          again := strings.ToLower(readLine("\nCalculate again? (y/n): "))
67          if again != "y" {
68              fmt.Println("Goodbye!")
69              break
70          }
71      }
72  }
```